# HOW TO Build A HTPC

**Combine the big screen of your TV and the processing power of your computer and get a Home Theatre Personal Computer**

**Bhaskar Sarma**

TVs and computers are the two most ubiquitous devices in our households, and each has its advantages and disadvantages. A TV concentrates on image and sound quality, but user control extends to just switching the thing on and off, flipping channels, and changing a setting or two. A computer, on the other hand, is all about user interactivity, but unless you have deep pockets, your PC screen will be significantly smaller than the TV you want. This means it cannot compare to the experience of watching a movie on a large screen.

If you build yourself a home theatre personal computer (HTPC), though, you can combine the best of the two, not to mention record TV programmes, rewind, fast forward, or pause a live TV broadcast, enjoy audio, and sometimes, live radio. Another practical aspect is that a HTPC saves space—the TV screen doubles up as a computer monitor when you need it to.

A HTPC means you'll need to buy both software and hardware; here's what you need.

## The Hardware

**STEP 1** Get yourself a case that doesn't have noisy fans—noise from the cooling fans can mar the experience. Also, use a case that is well-ventilated and looks good; you won't want an ugly-looking case messing up the décor of your living room. There are cases specially built for HTPC systems that fit these requirements. We recommend the iBall Engine-X (Rs 1,700). You might want to ask your vendor for a micro power supply from VIP or Cooler Master (around Rs 800), though they aren't easily available.

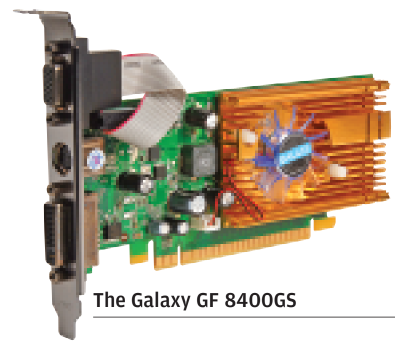
The iBall Engine-X

**STEP 2** The processor needs to be powerful enough to handle multithreaded applications like video encoding and decoding. Dual-cores are preferable. We recommend the AMD Athlon64 X2 5400+ (Rs 6000), 5600+ (Rs 6300), and the Intel Core 2 Duo E6550 (Rs 7,800).

**STEP 3** The motherboard need not be built for overclocking or have support for 45 nm processors. In fact, overclocking is the bane of HTPC systems; such motherboards emit digital noise, which affects picture and sound quality. It's preferable that these motherboards have passive cooling solutions like heat dispersing pipes to ensure quiet performance. HDMI slots for connecting to the TV are vital if you want to use the onboard graphics card. We recommend, for AMD processors, the MSI K9AGM2-690V (Rs 3,100) and the ASUS M2A-VM HDMI (Rs 5,225); for Intel processors, we'd say the Gigabyte GA- G33M S2H (Rs 5,950) and the Abit Fatalty F-190 HD (Rs 7,500).

The GigaByte-GA-G33M-S2H

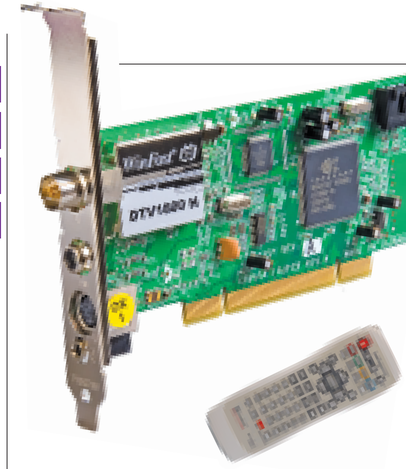
**STEP 4** The video card need not be too heavy-duty as the primary role is display, not serious gaming. The cards should have a low power requirement, good video decoding capabilities, and a small form factors. We recommend the Galaxy GF 8400GS (Rs 2,900).



The Galaxy GF 8400GS

**STEP 5** The TV-Tuner + video capture card is what distinguishes a HTPC from a regular PC. This card picks up live TV signals from either satellite or cable and outputs them live to the screen. The video capture part encodes the signals into a different format, usually MPEG2, and stores them for future viewing. The card could be single- or dual-tuner, and provide features like time-shifting (which means the ability to pause, rewind or fast-forward a live TV broadcast). Other formats into which video is stored are DivX and WMV.

Good-quality cards usually come with a software bundle and a remote. If you're using a direct-to-home setup, you'll need to buy a video card with composite inputs (the red, yellow and white connectors). We recommend the Leadtek Winfast DTV1800H (Rs 3,990).



The Leadtek Winfast DTV1800H

**STEP 6** It doesn't hurt to have some extra memory; 1 GB of RAM is sufficient, though with Vista, you're better off with 2 GB. Always use RAM in dual-channel mode—two 512 MB sticks if you want 1 GB of RAM; it's much faster than a single 1 GB stick. We recommend 2x 512 MB Kingston DDR2 667 MHz RAM (Rs 800 each), or 2 x 1 GB of the same (Rs 1,450 each).

**STEP 7** Both the hard drive and the optical drive must be as noiseless as possible. If you plan on recording and storing a lot of TV content—which would be natural, since it's a home theatre PC you're talking about—you'll need a sufficiently large hard drive. The optical drive should be a DVD-RW drive if you want to carry your content around. We recommend the WD Caviar SE16 WD3200AAKS (320 GB, Rs 4,500), and the Lite-On LH-20A1P (DVD-Writer, Rs 1,700).

The Lite-On LH-20A1P

**STEP 8** The mouse and keyboard needed to control the software should be wireless—so you can sit on a couch comfortably and control your system. The wireless combo should not be infrared, as line-of-sight issues crop up and are a pain to resolve. We recommend the Logitech Cordless Desktop MX 3200.

Logitech Cordless Desktop MX3200 (keyboard)

**STEP 9** The TV is the centrepiece of your home theatre system. By default, the TV needs to be high definition and classy. Check out the Diwali buying guide in last month's issue for more info on our recommended product, the Samsung Bordeaux LA32R81B (Rs 52,000).

The Samsung Bordeaux LA32R81B

**STEP 10** No home theatre experience is complete without the sound system.The TV will have speakers of its own, but the sound quality of any TV, no matter how good it looks, will be nothing to write home about. The big picture on the huge TV is complemented with the surround sound experience. Get yourself a 5.1 speaker system for that. We recommend the Artis S6600R / FM (Rs 8,500).

Artis S6600R / FM

## Software

After you've plugged in all the hardware, there are still some things to do before getting the popcorn. If you're running Windows Vista Home Premium or better, you'll have Windows Media Center at your disposal. Windows XP Media Center Edition is only sold with a PC attached, so don't bother with that. Windows-based software like Media Portal (on this month's *Entertainment* DVD) allows for a lot more customisation. You can also build your system around Linux using freeware programs like MythTV (find it on the *Entertainment* DVD). In fact, you can get specialised Linux distributions like Knopp Myth and Mythbuntu that comes with MythTV installed. Linux-based systems offer a lot of customisation, but you need to ensure all your hardware is supported.

## Satisfaction!

We built our system keeping in view value for money, but if you have cash to spare, go all out! HTPC building is a hobbyist activity, and a Google search brings up a wealth of related information with guides and reviews.

Do bear in mind what an HTPC is good for and when it doesn't make sense—you don't want to spend all that money and end up unhappy!

*bhaskar_sarma@thinkdigit.com*